

Homework 2

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Problem 1 Let $X \sim \text{Bin}(n, p)$ be a binomial random variable with parameters n and p . Recall that the p.m.f of X is given by

$$p(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n.$$

(a) Verify that

$$p(k+1) = \frac{n-k}{k+1} \cdot \frac{p}{1-p} \cdot p(k), \quad k = 0, 1, \dots, n-1.$$

- (b) Write an algorithm (or “pseudo code”) for the inverse transform method for generating X that utilizes the relation in part (a), in a way that you do not have to compute all the p.m.f. values, and also you only have to compute the c.d.f. values on the fly if needed.
- (c) Implement your answer in part (b) on a computer to generate 100 copies of $\text{Bin}(10, 2/3)$ and plot their distribution.

Solution:

(a) For $k = 0, 1, \dots, n-1$, we can write the p.m.f. explicitly as follows:

$$\begin{aligned} p(k+1) &= \binom{n}{k+1} p^{k+1} (1-p)^{n-k-1} \\ &= \frac{n!}{(k+1)!(n-k-1)!} p^{k+1} (1-p)^{n-k-1} \\ &= \frac{n!}{(k)!(n-k)!} p^k (1-p)^{n-k} \left\{ \frac{n-k}{k+1} \frac{p}{1-p} \right\} \\ &= p(k) \cdot \frac{n-k}{k+1} \cdot \frac{p}{1-p}. \end{aligned}$$

(b)

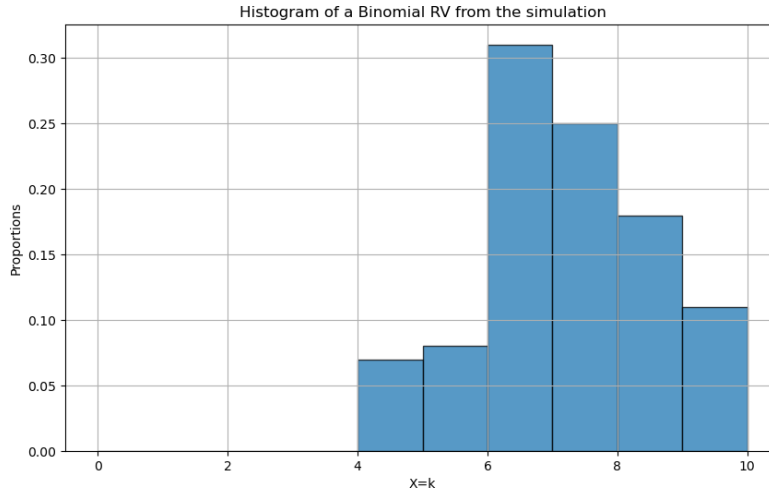
In the following algorithm, $prob$ and F represents $\mathbb{P}(X = i)$ and $\mathbb{P}(X \leq i)$ for a current value i , respectively. Therefore, this algorithm will terminate and return i when i is the smallest number in which $U \leq F(i)$. That is, the probability of getting i is $F(i) - F(i-1) = \mathbb{P}(X = i)$.

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1  $U \leftarrow \text{rand}(0, 1);$ 
2  $i \leftarrow 0;$ 
3  $prob \leftarrow (1-p)^n;$ 
4  $F \leftarrow prob;$ 
5 while  $F < U$  do
6    $prob \leftarrow \frac{(n-i)}{(i+1)} \cdot \frac{p}{1-p} \cdot prob;$ 
7    $F \leftarrow F + prob;$ 
8    $i \leftarrow i + 1;$ 
9 return  $i;$ 

```

(c) See the figure below.



Problem 2 In this problem we will generate a negative binomial random variable with parameters r, p in three different ways. This random variable, called $NB(r, p)$, is the number of independent Bernoulli trials needed to get r successes, where each Bernoulli trial has success probability p . It has a p.m.f. given by

$$p(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, k = r, r+1, \dots$$

- (a) Verify the relation

$$p(k+1) = \frac{k(1-p)}{k+1-r} p(k).$$

- (b) Use the relation in part (b) to give an algorithm for generating $NB(r, p)$.
- (c) Write down the relationship between $NB(r, p)$ and $Geom(p)$. Use it to obtain an algorithm to generate $NB(r, p)$.
- (d) Using the interpretation that $NB(r, p)$ counts the number of independent $Ber(p)$ trials required to accumulate r successes, obtain yet another approach for generating $NB(r, p)$.
- (e) Implement your algorithms in parts (a), (b) and (c) on a computer for generating $NB(2, 1/3)$. For each algorithm, generate 100 copies of X and plot their distribution. Are the plots similar?

Solution:

- (a) We can write the p.m.f. explicitly as follows:

$$\begin{aligned} p(k+1) &= \frac{(k)!}{(r-1)!(k+1-r)!} p^r (1-p)^{k-r+1} \\ &= \frac{k(1-p)}{k+1-r} \frac{(k-1)!}{(r-1)!(k-r)!} p^r (1-p)^{k-r} \\ &= \frac{k(1-p)}{k+1-r} p(k). \end{aligned}$$

(b) For a current value of k , $prob$ and F represents $P(X = k)$ and $P(X \leq k)$, respectively. Then, we update $P(X = k + 1)$ and $P(X \leq k + 1)$ using the relationship in (a). The algorithm is presented below.

```

1  $U \leftarrow \text{rand}(0, 1)$ ;
2  $k \leftarrow r$ ;
3  $prob \leftarrow p^r$ ;
4  $F \leftarrow prob$ ;
5 while  $F < U$  do
6    $prob \leftarrow \frac{k(1-p)}{(k+1-r)} \cdot prob$ ;
7    $F \leftarrow F + prob$ ;
8    $k \leftarrow k + 1$ ;
9 return  $k$ ;
```

(c)

We consider r independent random variables $Y_1, Y_2, \dots, Y_r$ that follows a geometric distribution with parameter p . We know that

$$Y_1 + Y_2 + \dots + Y_r \sim NB(r, p).$$

So to generate a random variable $NB(r, p)$, we'll generate r *i.i.d.* $Geom(p)$ random variables and add them together.

From Lecture 3, we note that if $U \sim Unif(0, 1)$, then

$$\left\lceil \frac{\log(1 - U)}{\log(1 - p)} \right\rceil \sim Geom(p).$$

So our procedure for generating $NB(r, p)$ is as follows: first, generate r *i.i.d.* $Unif(0, 1)$ random variables $U_1, \dots, U_r$. Next, create r *i.i.d.* $Geom(p)$ random variables using the transformation:

$$Y_i = \left\lceil \frac{\log(1 - U_i)}{\log(1 - p)} \right\rceil$$

for $i = 1, \dots, r$. Finally, return the sum $Y_1 + \dots + Y_r$

```

1  $X = 0$ ;
2 for  $i \leftarrow 1$  to  $r$  do
3   Generate  $U \sim Unif(0, 1)$ ;
4   Apply  $Y = \left\lceil \frac{\log(1-U)}{\log(1-p)} \right\rceil$ ;
5   Update  $X \leftarrow X + Y$ 
6 return  $X$ ;
```

(d)

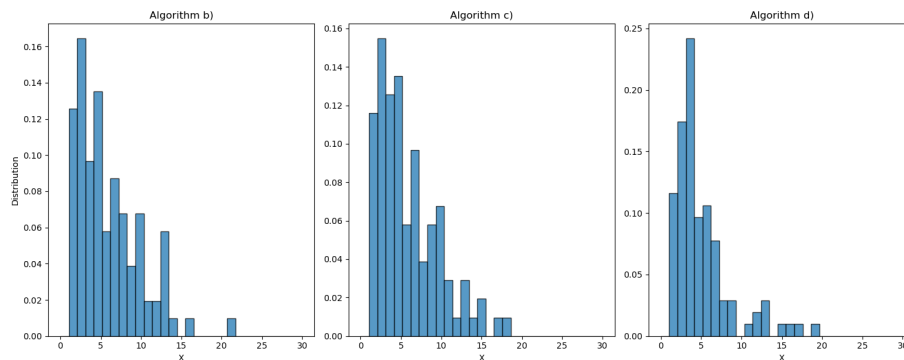
We can use the interpretation that $NB(r, p)$ counts the number of *i.i.d.* $Bern(p)$ trials required to accumulate r successes to generate $NB(r, p)$ random variables as follows: we sequentially generate *i.i.d.* $Bern(p)$ random variables and keep track of the number of successes and the number of trials in total as we generate them. After each iteration, we check to see if we have r successes. If we do, we terminate; if we don't, then we continue on. Pseudo code for generating a $NB(r, p)$ random variable in this way is below.

```

1  $n = 0$ ;
2  $s = 0$ ;
3 while  $s < r$  do
4   Generate  $B \sim \text{Bern}(p)$ ;
5    $s = s + B$ ;
6    $n = n + 1$ ;
7 return  $n$ ;

```

(e) These plots below are quite similar.



Problem 3 Write an algorithm to generate the random variable X where

$$\mathbb{P}(X = j) = \left(\frac{1}{2}\right)^{j+1} + \left(\frac{1}{2}\right) \frac{2^{j-1}}{3^j}, j = 1, 2, \dots$$

Implement your algorithm on a computer to generate 100 copies of X , and plot their distribution.

Solution:

We will implement an inverse transform method for this distribution without computing the entire p.m.f. upfront. For a current value j , we update F to be equal to $\mathbb{P}(X \leq j)$ using the following recursion:

$$\mathbb{P}(X \leq j + 1) = \mathbb{P}(X \leq j) + \mathbb{P}(X = j + 1) = \mathbb{P}(X \leq j) + \left(\frac{1}{2}\right)^{(j+1)+1} + \frac{1}{2} \frac{2^{(j+1)-1}}{3^{(j+1)}}.$$

This algorithm terminates and returns $j + 1$ when it encounters that

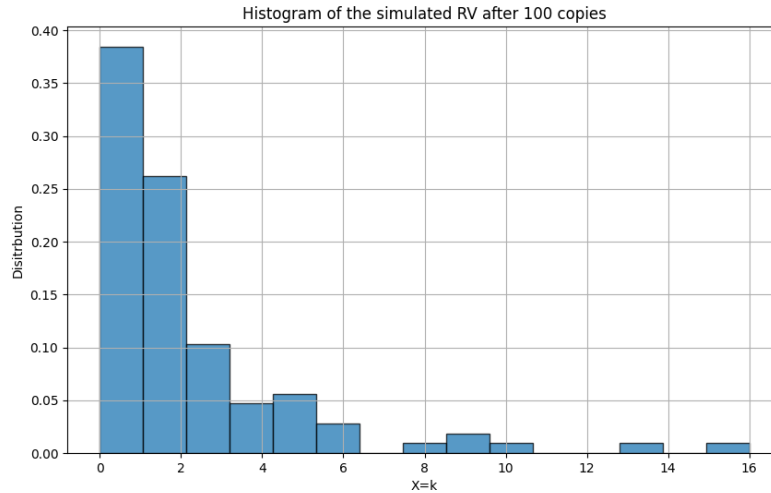
$$\mathbb{P}(X \leq j) < U \leq \mathbb{P}(X \leq j + 1).$$

```

1  $U \sim \text{Unif}(0, 1)$ ;
2  $j \leftarrow 1$ ;
3  $F \leftarrow \left(\frac{1}{2}\right)^2 + \frac{1}{2} \cdot \frac{1}{3}$ ;
4 while  $F < U$  do
5    $j \leftarrow j + 1$ ;
6    $F \leftarrow F + \left(\frac{1}{2}\right)^{j+1} + \frac{1}{2} \frac{2^{j-1}}{3^j}$ ;
7 return  $j$ ;

```

See the figure below.



Problem 4 A fair die is to be continually rolled until all possible outcomes $1, 2, \dots, 6$ have occurred at least once, and we are interested in the total number of rolls in this experiment. Implement an algorithm on a computer to generate 100 copies of the total number of rolls, and plot the distributions of these 100 copies.

Solution:

The general idea is, for each trial/copy, we keep rolling a die until all possible outcomes show up. Then, we record the total number of rolls in each trial. The algorithm is outlined below for $N = 100$ trials:

```

1  $N \leftarrow 100$ ;
2  $NumRolls \leftarrow []$ ;
3 for  $i \leftarrow 1$  to  $N$  do
4    $Seen \leftarrow []$ ;
5    $j \leftarrow 0$ ;
6   while  $length(Seen) < 6$  do
7     Generate  $X \sim Unif(1, 2, 3, 4, 5, 6)$ ;
8     if  $X \notin Seen$  then
9       Append  $X$  to  $Seen$ ;
10     $j \leftarrow j + 1$ ;
11  Append  $j$  to  $NumRolls$ ;
```

In each trial, we keep rolling a new dice from a uniform distribution over $\{1, 2, \dots, 6\}$ until we encounter all the faces. Then, we record this number j to 'NumRolls'. Then, we plot the distribution of the numbers of total rolls.

See the figure below for the 100 simulations.

